Can Agentic AI Settle a 20-Year-Old Open Problem?

The decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ — a machine-checked campaign

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GPT-5.4, 5.4 Pro, 5.5, 5.6 Pro, 5.6 Sol (OpenAI)

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The one-slide version

- In 2002/2003 I defined a family of spatial description logics and left their central decidability question open. It still is.
- In 2026 I directed a five-month AI campaign on it: frontier models, adversarial review, machine verification.
- The full question is not settled — but it is *reduced* to one well-posed lemma, with an exact arithmetical address (Π^0_1) on the way.
- A sub-question is settled: drop ∀PO and the logic is decidable, by a procedure the Lean kernel checks.

How to read everything in this talk

The mathematics is AI-produced and *unverified by human experts*. Standing label for the full logic: **strongly supported, *not* certified** — but the ∀PO-free fragment is **machine-certified**, reviewed three times.

Five ways two regions can relate (RCC5)

DR	PO	EQ	PP	PPI
disjoint	partial overlap	equal	proper part	proper superpart

- Every pair of regions stands in exactly one of these five relations.
- *Jointly exhaustive, pairwise disjoint* — the classic qualitative spatial calculus (a coarser cousin of RCC8).
- Think GIS, layout, biology, any "what contains / touches what" domain — no coordinates required.

Reasoning without geometry: the composition table

If I know $x\ R\ y$ and $y\ S\ z$, what can $x\ ?\ z$ be?

known	known	forced / allowed
x PP y	y PP z	x PP z (*forced*: parts of parts are parts)
x PP y	y DR z	x DR z (*forced*: inside something disjoint)
x DR y	y DR z	anything at all (*free*: 5 options)

- All 25 cells form the composition table — pure algebra, derived once from set semantics, then used forever.
- Abstract semantics: a "model" is any relation assignment closed under this table — *space itself has left the building*.
- Every probe in this project re-derives the table from scratch, so the algebra can never silently drift.

A description logic: you write definitions and facts

A TBox is a set of concept *definitions*, built from properties and quantified spatial relations:

$\mathsf{CityAtRiver} \equiv \mathsf{City} \sqcap \exists \mathsf{PO}.\,\mathsf{River}$		a city *overlapping* a river
$\mathsf{GermanCity} \equiv \mathsf{City} \sqcap \forall \mathsf{PP}.(\neg \mathsf{Country} \sqcup \mathsf{Germany})$		a city in *no country but* Germany

Then we assert a few primitive facts — and nothing more:

$$\mathsf{Hamburg} \sqsubseteq \mathsf{City} \sqcap \exists \mathsf{PO}.\,\mathsf{Alster} \qquad \text{(a city, overlapping the Alster)}$$

$$\mathsf{Alster} \sqsubseteq \mathsf{River} \sqcap \exists \mathsf{PP}.\mathsf{Germany} \sqcap \forall \mathsf{PO}.\neg \mathsf{Country} \sqcap \forall \mathsf{PP}.(\neg \mathsf{Country} \sqcup \mathsf{Germany})$$

Note what we did *not* say: that Hamburg is a GermanCity, or a CityAtRiver.

. . . and the reasoner *infers* the classification

We never *asserted* that Hamburg is a GermanCity or a CityAtRiver — the reasoner derives both.

City at a river (immediate):

$$\text{Hamburg PO Alster} \wedge \text{Alster} \sqsubseteq \text{River} \implies \text{Hamburg} \sqsubseteq \text{City} \sqcap \exists \text{PO.River} = \text{CityAtRiver}.$$

German city — needs the composition table. For any country c with Hamburg PP c:

- Alster PO Hamburg PP c $\Rightarrow$ Alster $\{\text{PO}, \text{PP}\}$ c $\qquad$ $\text{comp}(\text{PO}, \text{PP}) = \{\text{PO}, \text{PP}\}$
- Alster overlaps no country ($\forall \text{PO}.\neg\text{Country}$) $\Rightarrow$ Alster PP c
- Alster's only country-superpart is Germany $\Rightarrow$ c = Germany

So Hamburg *is* a German city — and every hidden fact reduces to concept satisfiability: $C \sqsubseteq D$ iff $C \sqcap \neg D$ unsat.

$\mathcal{ALCI}_{\mathrm{RCC5}}$: the spatial relations *become* the roles

The logic (Wessel 2002/2003):

- $\mathcal{ALCI}$ = the basic DL with inverse roles;
- roles = the five RCC5 relations themselves;
- the composition table is enforced globally;
- EQ = identity (*strong* semantics).

So one can say things like

$$\forall \mathsf{PP}.\, \neg \mathsf{Lake} \qquad \exists \mathsf{DR}.\, \mathsf{Reserve}$$

"no part is a lake", "disjoint from some reserve".

Why this is different
Concrete-domain DLs constrain spatial *attributes* of objects. Here the relations are *first-class roles*: one can quantify over them. None of the known concrete-domain decidability results apply.

Why the standard toolbox dies on contact

Every pair of elements carries a relation $\Rightarrow$ a model is a complete labelled graph, never a tree.

no tree models

Blocking/unravelling — the workhorse of DL decidability — has no tree to work on.

no finite models

$\exists \mathrm{PPI}.\top \sqcap \forall \mathrm{PPI}.\exists \mathrm{PPI}.\top$ forces an infinite ascending part-of tower.

nondeterministic table

Most cells allow several outcomes — composition *constrains* but rarely *determines*.

Creating one new element forces relations to *every* existing element — and a dormant $\forall$ anywhere may wake up because of it.

The landscape: open exactly at the top

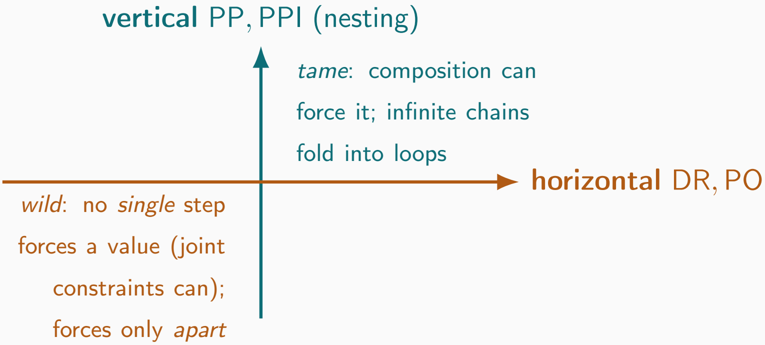
Logic	Relations	Concept satisfiability
$\mathcal{ALCI}_{\mathrm{RCC1}}$	$\{\mathsf{SR}\}$	decidable ($\equiv$ S5)
$\mathcal{ALCI}_{\mathrm{RCC2}}$	$\{\mathsf{DR}, \mathsf{O}\}$	decidable
$\mathcal{ALCI}_{\mathrm{RCC3}}$	$\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$	decidable (via two-variable FO!)
$\mathcal{ALCI}_{\mathrm{RCC5}}$, $\forall$PO-free	RCC5 w/o $\forall$PO	**decidable** (this project)
$\mathcal{ALCI}_{\mathrm{RCC5}}$	all five	**open**; PSpace-hard
$\mathcal{ALCI}_{\mathrm{RCC8}}$	eight RCC8 rel.	**open**; ExpTime-hard

Lutz & Wolter 2006, after proving the *topological* cousins undecidable:

> *"Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [$=\mathcal{ALCI}_{\mathrm{RCC5}}$] … to which the reduction exhibited in Section 8 does not apply."*

Recognized open by the leading experts — for twenty years.

One logic, two completely different axes



- Depth (nesting) is never the problem — ordinary modal-logic folding handles it.
- Everything hard lives in horizontal crowds: many pairwise-distinct neighbours that no finite summary seems to capture.

The asymmetry, made precise (machine-verified)

No *single* step forces a horizontal value. Of the 16 compositions of proper relations, exactly four return a single value:

$$\mathrm{comp}(\mathrm{DR},\mathrm{PPI}) = \mathrm{comp}(\mathrm{PP},\mathrm{DR}) = \{\mathrm{DR}\},\quad \mathrm{comp}(\mathrm{PP},\mathrm{PP}) = \{\mathrm{PP}\},\quad \mathrm{comp}(\mathrm{PPI},\mathrm{PPI}) = \{\mathrm{PPI}\}$$

— every one runs through PP/PPI; no single cell yields $\{\mathrm{PO}\}$. But the *intersection* of several can force PO ($\mathrm{comp}(\mathrm{DR},\mathrm{PO}) \cap \mathrm{comp}(\mathrm{PPI},\mathrm{PO}) = \{\mathrm{PO}\}$) — so "shadow" is joint closure, not single-path.

Disjointness is half-deterministic.

downward: $x\,\mathrm{DR}\,y \Rightarrow$ every part of x is DR from every part of y

upward: a superpart of x may be DR, PO, or PPI to y — free

So one high DR edge dispatches an *arbitrarily wide* crowd of disjointnesses below it, for free. What resists: PO, and the upward direction.

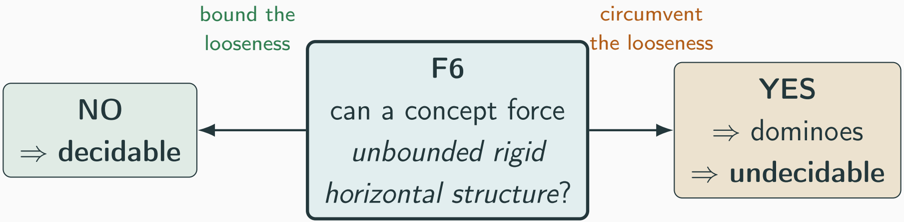
Shadows, live width, and the one property that matters

- An edge is a shadow if composition already forces it from edges you have — it costs a decision procedure *nothing*.
- The live width of an interface = the number of *non-shadow* edges there: the genuine branching that must be recorded.

Property F6 (the keystone of everything)
Is the live width of models bounded by a computable function of the input concept?

If yes $\Rightarrow$ finite certificate space $\Rightarrow$ satisfiability is decidable.

The knife's edge: one fact, two futures



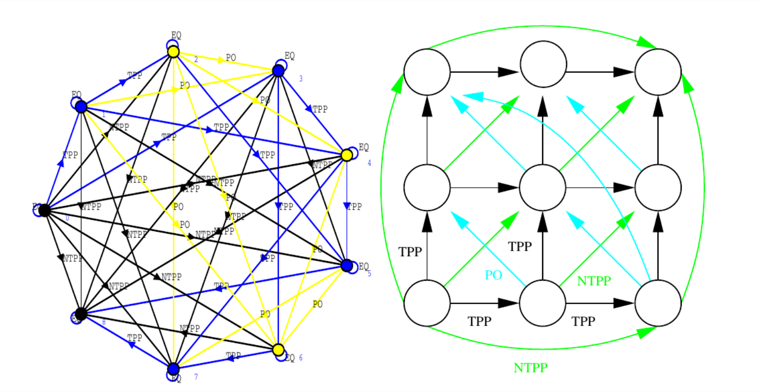
The *same* looseness of the composition table blocks both proofs. That is *why* neither direction has been settled in twenty years — it is one question seen from two sides.

The standard weapon needs a rigid grid

- Classic undecidability proofs encode domino tilings of $\mathbb{N} \times \mathbb{N}$: build a grid, force tile constraints, reduce the halting problem.
- The crux is rigidity: going *east then north* must land on *exactly the same cell* as *north then east* ($R_X \circ R_Y = R_Y \circ R_X$, the "coincidence condition").
- Every known route into $\mathcal{ALCI}_{\mathrm{RCC}}$ needs a feature it does not have: number restrictions, arbitrary role axioms, topological rigidity, nominals...

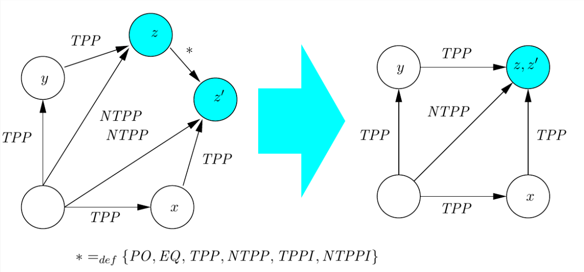
Question (2003 and today): can the composition table itself supply the rigidity?

2003: the grid models *exist*. . .



An RCC8 network isomorphic to an $n \times n$ grid (report7, Fig. 10, 2002/2003) — the construction *found by a Lisp enumerator* and drawn with CLIM (MCL), not by hand. The models are *out there*.

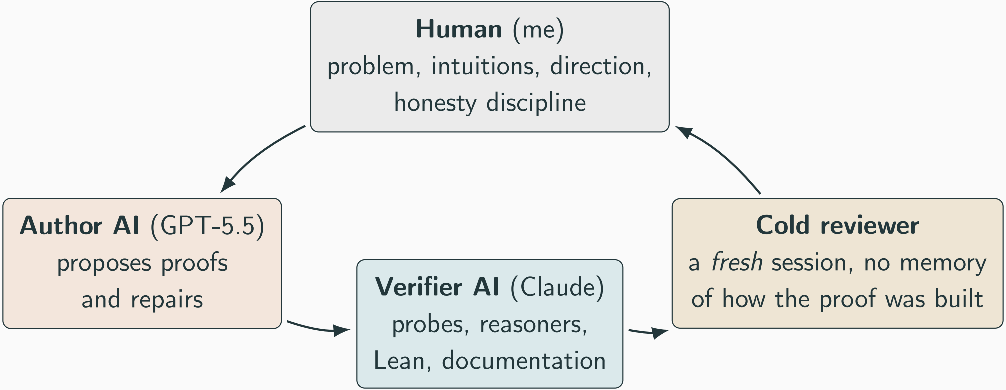
…but no concept can *force* them: the coincidence obstruction



report7, Fig. 11: successors z (east-north) and z' (north-east) *should* be one corner — the table *alone* allows six relations, forcing neither merge nor split. You *can* force the merge (clash out the five non-EQ options, EQ remains) — but that's the trap: those markers collapse *every* corner at the next level (SAT at level 1, UNSAT at 2; RCC5 *and* RCC8). Neither gives a grid: loose ⇒ no join, forced ⇒ total collapse.

> *"Even though these infinite models do exist, we have found no way to enforce the depicted model(s) by means of a concept term."* *(report7, 2003)*

The cast, and the loop



Propose → verify → attack → repair. Round after round, March–July 2026.

The campaign at a glance

- ∼4 months, ∼30 repair rounds
- 20 adversarial reviews — a defect or overclaim found in *all but two*
- 88 self-contained verification probes (each re-derives the algebra from set semantics)
- 4 Lean files, ∼5,900 lines, zero `sorry`
- 2 working reasoners, cross-validated on 911 concepts

The single most telling number
Across *all* rounds and reviews, no accepted-but-unsatisfiable concept was ever found. Every defect was a gap in an *argument* — never a counterexample to the intended theorem.

That is why the label is *strongly supported* — and why it is still *not certified*.

Route 1: tableaux with blocking — broken by totality

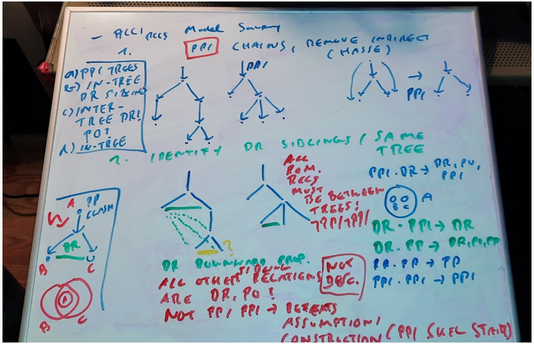
The default DL method: build a model top-down, *block* when a node repeats an ancestor's type.

- Here, composition forces edges into already-built nodes: from $x\,\mathrm{PP}\,y$, $y\,\mathrm{DR}\,z$ the table forces $x\,\mathrm{DR}\,z$.
- Diagnostic: $C_{\mathrm{force}} = \exists \mathrm{PP}.(\exists \mathrm{DR}.A) \sqcap \forall \mathrm{DR}.\neg A$ is **unsatisfiable** — any procedure that ignores forced edges wrongly accepts it.
- Dormant $\forall$'s reawaken blocked nodes; no fixed-arity local summary (types, triangles, quadruples, profiles, ...) is ever enough.

Lesson: in a total-relation logic, "local" does not exist.

What may be reused depends on the whole graph. (That is F6 in disguise.)

Route 2: the split-forest idea (a whiteboard, April 2026)



- Present the complete graph as a *forest of part-of trees*: split shared parts into EQ-mates, re-orient vertical edges into the trees.
- All remaining cross-edges are horizontal; complete them by the RCC5 *patchwork property* (a citable classical theorem).
- The one component every cold review found sound — the semantic foundation all later routes consume.

The human contribution: this decomposition. The models made it precise.

Route 3: tree automata — “acceptance without validity”

Package the search as a parity tree automaton (emptiness is decidable)...

- ...but the input tree omits the relations between fresh witnesses and the sources of $\forall$-constraints.
- An *alternating* automaton explores in independent copies — and copies guess the missing relations independently.
- A cold review built the killer: two copies privately guess $\{\mathrm{PP}\}$ and $\{\mathrm{DR}\}$ for the same pair; $\mathrm{comp}(\mathrm{PP},\mathrm{PP}) \cap \mathrm{comp}(\mathrm{PP},\mathrm{DR}) = \{\mathrm{PP}\} \cap \{\mathrm{DR}\} = \emptyset$ — every copy accepts, no model exists.

So the automaton was abandoned, in favour of explicit finite certificates checked by composition tests — with a *truth lemma* instead of an acceptance condition. (This is what got formalized.)

Machine-checking: what the Lean kernel buys

- Lean 4: a proof assistant whose kernel type-checks *every* inference; a proof with a gap *does not compile*.
- Our discipline: Lean 4 core only — no libraries; every object built from first principles; tiny trusted base.
- "`zero sorry`" = no admitted gaps anywhere.

Why we moved to Lean
Most reviews broke a *prose* proof — typically at a step that was *assumed*, not proved. The kernel cannot be talked past. After the move, reviews stopped finding soundness defects and started finding only *interface* gaps — which were then closed.

What exactly is certified (zero `sorry`)

Result	Meaning	File
Soundness pipeline	valid certificate $\Rightarrow$ real RCC5 model	`Round19Transport`
Abstraction faithful	satisfiable $\Leftrightarrow$ Hintikka-realizable	`Round19Transport`
RCC5 normal form (both dir.)	network $\Leftrightarrow$ order + disjointness	`RCC5NormalForm`
Forcing reduction	F6 fails $\Leftrightarrow$ a concept forces crowds	`ForcingReduction`
Dovetail schema	qualitative F6 $\Rightarrow$ decidable	`SemiDecidability`
$\forall$PO-**free decided** (2026-08-29)	a *decision procedure*, raw input	`POFreeLift`
Abstract $=$ **concrete**	composition table $\Leftrightarrow$ real sets	`POFreeLift`

What is deliberately not in the kernel: the open completeness premise itself (F6). It is *stated*, never axiomatized — the honest boundary between what is proved and what is conjectured.

The main theorem (conditional, soundness certified)

Conditional decidability
If F6 holds — live width bounded by a computable function of the concept — **then** $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability is decidable.

- The soundness half is kernel-checked end to end: certificate syntax $\to$ closure conditions $\to$ proper RCC5 frame $\to$ truth lemma $\to$ *actual model of the input concept*.
- The certificate checker is a real executable Boolean function — no oracle anywhere (a previous version had a hidden one; a cold review caught it; it was rebuilt).
- Everything that remains open is the antecedent — F6.

Unconditional win #1: a decidable fragment — now *certified*

Theorem ($\forall$PO-free fragment), machine-checked 2026-08-29
Concepts with no $\forall$PO subformula are decidable. In Lean, with fragment membership the *only* hypothesis: `decidableFSat` (raw input) `decidableSat_cone` (NNF) `decidableSetSat` (real sets)

- Still allowed: $\exists$PO, $\forall$DR, $\exists$DR and all part-of modalities — an expressive spatial fragment.
- On raw input the condition is *polarity-sensitive*: no $\forall$PO positively, no $\exists$PO *negatively*.
- Reached three times by three routes; the third (GPT-5.6 Sol's *cone scheme*) is the one that certified.
- Decidable $\neq$ runnable — doubly exponential.

Why *that* fragment — the one-sentence reason

The residual default
When composition leaves a pair's relation open, you can always settle it as PO — *unless some formula can object to a* PO *edge*. The only formula that can is $\forall \mathrm{PO}.D$.

- So in the fragment the lazy arrangement — declare only what a birth forces, default every remaining pair to PO — is always a model.
- Hence nothing ever has to be coordinated: every relation between two different children is the residual, and the residual is free. That is why the construction needs no joint realizability.
- Put $\forall$PO back and you do not lose one case of one induction — you lose the default itself. That is the whole boundary.

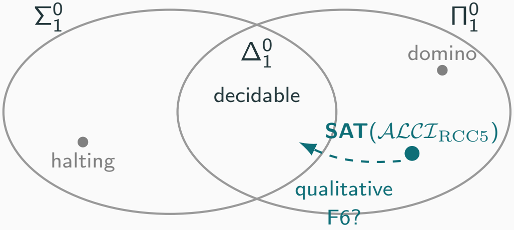
Unconditional win #2: the RCC5 normal form

Theorem (both directions kernel-checked, arbitrary domains)
A strong-EQ RCC5 network is composition-closed *iff* it is an ordered-disjoint structure:

- PP is a strict partial order (a genuine part-of hierarchy);
- DR is a symmetric, downward-closed disjointness;
- comparable pairs never disjoint; PO is the residual.

Every RCC5 model, anywhere, *is* just: a hierarchy + a downward-inherited "keeps-apart" relation + overlap as the leftover. The "$\Rightarrow$" is a 150-line kernel proof (`propext` only); the "$\Leftarrow$" is GPT-5.6 Pro's canonical set representation, also kernel-checked for arbitrary domains — the structural fact both major proof routes silently relied on, now a certified biconditional.

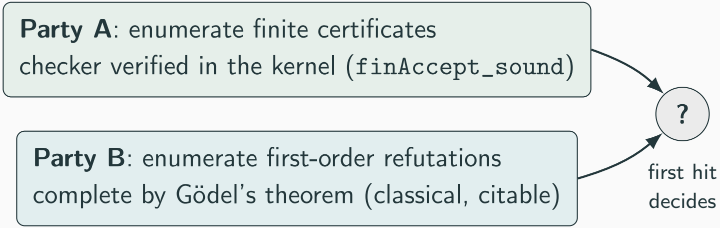
A late surprise: the problem's exact address is Π^0_1



- The whole semantics is finitely first-order axiomatizable; by Gödel completeness, *un*satisfiability is enumerable.
- So SAT is co-r.e. (Π^0_1) — the domino problem's *ceiling* (membership; we don't claim hardness).
- Nobody in the project — or the reviews — had noticed for four months. Machine-cross-checked; the reduction is in Lean.

Consequence: half of the keystone was never needed

A Π^0_1 problem is decidable *iff* SAT is also enumerable. So run two search parties and wait for the first hit:



- Needed now: only qualitative F6 — *some* finite certificate per satisfiable concept, no computable bound.
- The bound the reviewers kept demanding (and that kept breaking) comes back for free, afterwards — a theorem of the schema.

“Isn’t this just first-order logic?” — yes, and it helps

$\mathcal{ALCI}_{\mathrm{RCC5}}$-SAT *is* FO satisfiability of a finite theory. Does it land in a decidable fragment of FO?

Fragment	Fate	Why
two-variable FO^2	$\times$	composition is ternary (RCC3 *did* fall this way in 2003!)
prefix classes	$\times$	unbounded $\forall\exists$ alternation, FMP
guarded / clique-guarded	$\times$	totality is the one unguardable axiom
unary/guarded negation	$\times$	finite model property; totality again
guarded + transitivity	$\times$	undecidable outside guards

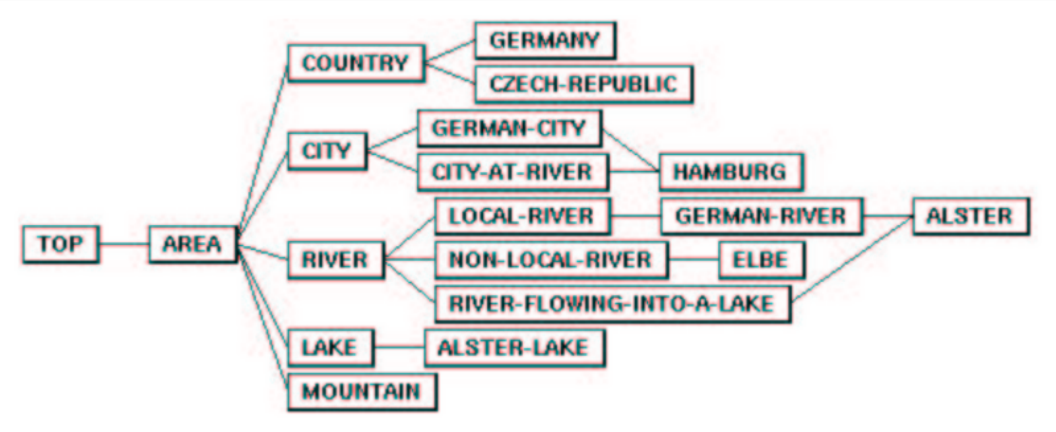
Every failure traces to the same fact: every pair is related, so models are maximally non-local — exactly the obstruction F6 names. The FO view rediscovers the project’s wall from classical model theory.

A working prototype reasoner (validated, not proved)

- The cover-tree tableau ($\sim$350 lines of Python): exact-occurrence blocking on the split-forest foundation.
- Cross-validated against an independent oracle on 911 concepts — zero mismatches; 775/775 structured decomposition tests; never once contradicted in the whole campaign.
- Decides correctly even the known trap (the strong-EQ PO-loop) that fools type-based procedures.

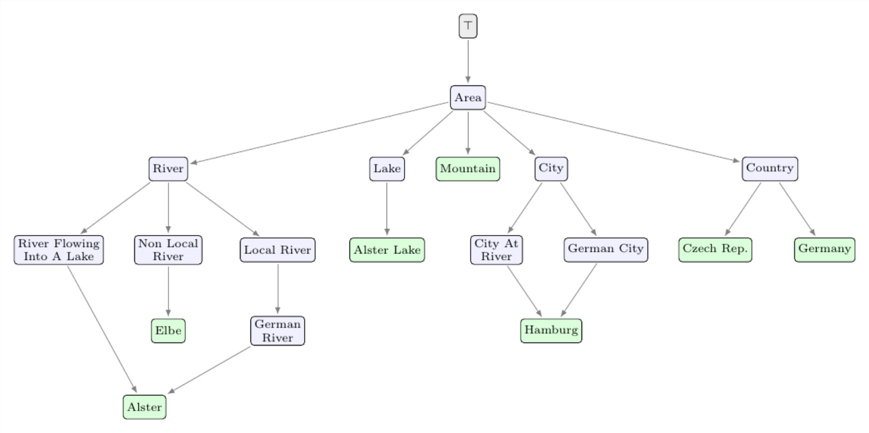
A running algorithm and a kernel-checked proof, pointing the same way — the strongest evidence short of a certificate.

2003: a prototype reasoner



The GIS concept taxonomy in report7 (Fig. 6), 2003 — itself *computed by a prototype reasoner* of the era, not drawn by hand.

2026, the cover-tree reasoner: all 21/21 — edge for edge



The 2026 cover-tree tableau reproduces it exactly — two reasoners, 23 years apart, identical result. (Green = individuals; *Hamburg* and *Alster* each have two parents — a DAG, not a tree.)

What the models did — and what they could not

Demonstrated

- months-long rigorous formal development, ∼30 rounds
- genuinely useful *adversarial* review (with machine-checked witnesses)
- a 5,900-line kernel-checked Lean development
- working, validated software

Not demonstrated

- the decisive novel idea — F6 is uncracked
- reliable *self*-review (both models declared victory prematurely; both were caught)
- working unsupervised: the human directed every round

Neither model is self-sufficient. What carried the project was the loop, not any single model.

The method that worked (steal this)

1. **Cold review beats hot review.** A *fresh* session with no memory of the construction found what warm reviewers missed — every time it mattered. Make it a standard step.
2. **Don't trust — verify.** Every load-bearing claim was cross-checked mechanically: 88 probes that re-derive the algebra from set semantics, and ultimately the Lean kernel. The machine checks are the ground truth, not the model's confidence.
3. **Label honestly.** *Strongly supported* $\neq$ *certified*. Overclaiming happened twice; cold review caught both; the claims were withdrawn — on the record.

The full timestamped conversation log is public — every idea, defect, and repair traceable to its origin, human or model.

If nothing else: a message to fellow researchers

- Whatever the fate of the mathematics, this campaign is an existence proof of what frontier models could already do in serious formal research — *in July 2026*.
- Anyone doing serious logic or mathematics should know these capabilities exist. Ignoring them does not advance science; declining to use them is simply unwise.
- Saying exactly this was a stated purpose of the work. Its first outing (DL 2026) was rejected without engaging either the mathematics or the message — so it is now on arXiv, in the open, with the full audit trail.

The instruments have changed. The standards — verification, adversarial review, honest labels — matter more than ever.

The results, by verification level

Level	Results	What it means
Lean-certified	soundness pipeline; abstraction faithfulness; RCC5 normal form (both directions); ∀PO-free lift core; F6 $\Rightarrow$ decidable; Π^0_1 dovetail	kernel-checked, with a stated axiom footprint
Theorem-level	∀PO-free decidability + bound; identity-selector characterization; W2′ folds into F6	proof in a work package, often with probes
Empirical	cover-tree tableau; 911-concept cross-validation; model verifier	test evidence, not a decision theorem
Open	static F6; completeness of the certificate scheme; automatic-cover property; full decidability	no proof claimed

Four levels kept rigorously distinct — conflating them is the easiest way to misread the project.

Where it stands

One well-posed question

can a concept force unbounded *identity-selector* structure?

NO $\Rightarrow$ decidable (machinery ready, kernel-checked) YES $\Rightarrow$ the grid/undecidability route opens

- 20+ years of difficulty, compressed into one lemma — with certified machinery waiting on both sides of the answer.
- And one sub-problem removed from the pile: drop $\forall$PO and it is decided, in the kernel, on raw input, under real set semantics — three cold reviews, no counterexample.
- Everything reproducible: papers, 4 Lean files (>45k lines, zero `sorry`), 100+ probes, 2 reasoners, full audit log.

Open since 2003. Smaller than ever. Still standing —

Backup: the full RCC5 composition table

comp	$\mathrm{DR}(a,b)$	$\mathrm{PO}(a,b)$	$\mathrm{EQ}(a,b)$	$\mathrm{PPI}(a,b)$	$\mathrm{PP}(a,b)$
$\mathrm{DR}(b,c)$	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
$\mathrm{PO}(b,c)$	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
$\mathrm{EQ}(b,c)$	DR	PO	EQ	PPI	PP
$\mathrm{PP}(b,c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$\mathrm{PPI}(b,c)$	DR	DR, PO, PPI	PPI	PPI	$*$

Row $= S(b,c)$, column $= R(a,b)$, entry = possibilities for (a,c); $* =$ all five. Re-derived from finite set semantics by every probe; matches the certified Lean table cell for cell.

Backup: what a finite certificate looks like

- **Catalogue** (the finite blueprint):
  - a *core network* — the non-repeating part of the model;
  - finitely many *attachment templates* — recipes for gluing a fresh witness cluster onto an interface;
  - a *steering function* fixing the horizontal relations from old occurrences to fresh ones (the only real freedom).
- **Plan**: which template to apply where; *unfolding* it (possibly forever) produces the split-forest model.
- **Soundness**, kernel-checked: one finite check on the catalogue guarantees *every* unfolding is a legal RCC5 model.
- **Live width** = live occurrences that must coexist at one interface during unfolding — the quantity F6 must bound.

Backup: the sharpened F6 — identity-selector minors

- The naive ways to force width (shell half-graphs, prefix pumps, PO-gaps) all collapse to finite catalogues — machine-checked.
- The sharp obstruction: n pairs (L_i, U_i), each kept apart by a private selector A_i and by nothing else — the selector incidence matrix is the $n \times n$ *identity*.
- A finite catalogue can hold only boundedly many private selectors, so: **F6 fails iff some concept forces such minors for every n.**
- Bare RCC5 networks with unbounded such structure *exist* (machine-checked) — so any F6 proof must use $\mathcal{ALCI}_{\mathrm{RCC}}$-*forceability*, not RCC5 consistency alone. The uniformization side-condition (W2′) provably folds into F6.